

Antipodes and Incidence Coalgebras*

WILLIAM R. SCHMITT

*Department of Mathematics, Massachusetts Institute of Technology,
Cambridge, Massachusetts 02139*

Communicated by Gian-Carlo Rota

Received October 10, 1986

We introduce the notion of a reduced incidence coalgebra of a family of locally finite partially ordered sets and describe how bialgebras and Hopf algebras arise as such coalgebras. In the case when a reduced incidence coalgebra is a Hopf algebra, we give explicit recursive and closed formulas for the antipode, each of which generalizes a corresponding formula from the theory of Möbius functions. We give applications to the inversion of ordinary formal power series and inversion of Dirichlet series. We also formulate the classical Lagrange inversion formula as a combinatorial formula for the antipode of a Hopf algebra arising from the family of finite partition lattices. © 1987 Academic Press, Inc.

INTRODUCTION

The theory of incidence algebras and reduced incidence algebras of partially ordered sets provides a comprehensive formal framework for the study of generating functions arising in enumeration theory. Not only can one obtain all the classical examples of generating functions (ordinary, exponential, Dirichlet and Eulerian, etc.) as reduced incidence algebras, but one is also given a systematic technique for setting up algebras of generating functions, or generalized formal power series, which are appropriate for given specific classes of enumeration problems.

It was pointed out by Goldman and Rota [6] and later by Joni and Rota [7] that incidence and reduced incidence coalgebras of partially ordered sets are much more basic objects than their duals, the incidence and reduced incidence algebras; this is due to the fact that underlying combinatorial structures can be immediately cast in coalgebraic terms. The combinatorial content of identities in incidence algebras often becomes apparent as soon as they are translated into dual, coalgebraic form.

* The author's Doctoral dissertation, Massachusetts Institute of Technology, Department of Mathematics, 1986. The author would like to thank his thesis supervisor, Professor Gian-Carlo Rota, for providing the inspiration for this work.

It was also recognized in [7] that certain reduced incidence coalgebras are endowed with compatible algebra structures, in the sense that they are actually bialgebras; and certain semigroups of generating functions could be obtained as the semigroups of multiplicative functions on these bialgebras. It was shown by Doubilet [3] that one such bialgebra, arising from an infinite lattice of partitions, actually could be embedded in a Hopf algebra. This embedding, however, was done in a purely algebraic manner, and the resulting Hopf algebra was not given a combinatorial interpretation as an incidence coalgebra.

In the present work we provide a formal setting in which one can always obtain generating functions as semigroups of multiplicative functions on incidence bialgebras. The subgroups of those generating functions consisting of invertible elements are then given as groups of multiplicative functions on corresponding Hopf algebras. We then have the necessary structure, namely the antipode of the Hopf algebra, in order to calculate the inverse of a given generating function in a purely combinatorial manner.

After introducing some basic definitions and results from the theory of Hopf algebras, we begin in Section I by giving an order-theoretic definition of an order compatible equivalence relation and proving its equivalence with the older, less intuitive, definition of order compatibility (see [4, 7, 9]). We then define the reduced incidence coalgebra of a family of locally finite partially ordered sets modulo an order compatible equivalence relation. Then we describe which semigroup structures on the set of equivalence classes, or types, of segments induce a multiplication which makes the reduced incidence coalgebra into a bialgebra.

In the second section we provide a simple answer to the question of when the standard reduced incidence coalgebra (i.e., reduced modulo the isomorphism relation) of a family of partially ordered sets is a bialgebra. This question was raised by Joni and Rota [7], who proved that for certain classes of matroids one obtains bialgebras as standard reduced incidence coalgebras. Our answer is very general; in fact, it turns out that none of the special properties of matroids are relevant to this question.

In Section III we give a combinatorial method for embedding a reduced incidence coalgebra into an incidence bialgebra, and show how, by this procedure, one can always obtain generating functions as semigroups of multiplicative functions on bialgebras.

In the fourth section we characterize those incidence bialgebras which are actually Hopf algebras and give an explicit recursive formula for the antipode of such a Hopf algebra. After showing how incidence bialgebras can be embedded in incidence Hopf algebras by a combinatorial construction, we arrive at our main result in section VI: a closed formula for the antipode of a Hopf algebra which is a reduced incidence coalgebra. This

formula, given as an alternating sum over chains in a segment, bears a striking resemblance to Phillip Hall's formula for the Möbius function of a poset.

In the final section we give a purely combinatorial formulation of the classical Lagrange formula for the compositional inverse of a formal power series. This is done by interpreting the Lagrange formula as a formula for the antipode of a Hopf algebra arising from the family of all finite partition lattices.

The analogy between the theory of Möbius functions (see [8]) and the theory of antipodes developed thus far is so compelling that we believe it is inevitable that more refined formulas for antipodes will be found, each having Möbius-theoretic analogues; and classical Lagrange inversion will follow as a special case of one (or more) of these formulas when applied to finite partition lattices.

0. BACKGROUND AND TERMINOLOGY

If P is a partially ordered set (poset) and $x \leq y$ in P then the *segment* $[x, y]$ is the poset consisting of all z in P such that $x \leq z \leq y$, with the ordering relation inherited from P . If Q is a segment then the unique minimal element of Q will be denoted by $\hat{0}$ and the unique maximal element by $\hat{1}$. A poset is said to be *locally finite* if every segment contained in it is finite. All posets considered in this work will be assumed locally finite.

A *coalgebra* is a triple (C, Δ, ε) with C a K -vector space (where K is some fixed field of characteristic 0), $\Delta: C \rightarrow C \otimes C$ a map called the *diagonalization*, and $\varepsilon: C \rightarrow K$ a map called the *counit*, where Δ and ε satisfy the commutative diagrams

$$\begin{array}{ccc}
 C & \xrightarrow{\Delta} & C \otimes C \\
 \Delta \downarrow & & \downarrow \Delta \otimes I \\
 C \otimes C & \xrightarrow{I \otimes \Delta} & C \otimes C \otimes C
 \end{array} \quad (\text{coassociativity})$$

and

$$\begin{array}{ccccc}
 & & C & & \\
 & \swarrow & & \searrow & \\
 K \otimes C & & & & C \otimes K \\
 & \nwarrow & \Delta \downarrow & \nearrow & \\
 & & C \otimes C & &
 \end{array} \quad (\text{counitary property}).$$

$\varepsilon \otimes I$ (left arrow) $I \otimes \varepsilon$ (right arrow)

We will usually write "the coalgebra C " instead of "the coalgebra (C, Δ, ε) ."

A *subcoalgebra* of a coalgebra C is a subspace W such that $\Delta(W) \subseteq W \otimes W$. A *coideal* of C is a subspace J such that $\Delta(J) \subseteq J \otimes C + C \otimes J$ and $\varepsilon(J) = 0$. A *coalgebra map* from (C, Δ, ε) to $(C', \Delta', \varepsilon')$ is a linear map $f: C \rightarrow C'$ satisfying the commutative diagrams

$$\begin{array}{ccc} C & \xrightarrow{f} & C' \\ \Delta \downarrow & & \downarrow \Delta' \\ C \otimes C & \xrightarrow{f \otimes f} & C' \otimes C' \end{array}$$

and

$$\begin{array}{ccc} C & \xrightarrow{f} & C' \\ \varepsilon \searrow & & \swarrow \varepsilon' \\ & K & \end{array}$$

THEOREM 0.1. *Let C be a coalgebra, J a coideal and $\pi: C \rightarrow C/J$ the natural linear map onto the quotient vector space. Then*

- (a) C/J has a unique coalgebra structure such that π is a coalgebra map,
- (b) if $f: C \rightarrow D$ is any coalgebra map then $\text{Ker } f$ is a coideal, and
- (c) if $J \subseteq \text{Ker } f$ then there is a unique coalgebra map $\bar{f}$ such that

$$\begin{array}{ccc} C & \xrightarrow{f} & D \\ \pi \searrow & & \swarrow \bar{f} \\ & C/J & \end{array}$$

is commutative.

A space B which is simultaneously an algebra and a coalgebra is said to be a *bialgebra* if Δ and ε are algebra maps. If C is a coalgebra and A is an algebra with multiplication $\mu_A: A \otimes A \rightarrow A$ then we give $\text{Hom}(C, A)$ an algebra structure by defining the product or *convolution* $f * g$ of f and g by

$$f * g = \mu_A \circ (f \otimes g) \circ \Delta. \quad (0.2)$$

The unit of this algebra is $u\varepsilon$, when u is the unit of A . The *dual algebra* C^* of a coalgebra C is the algebra $\text{Hom}(C, K)$.

PROPOSITION 0.3. *If B is a bialgebra and A is a commutative algebra then the set of algebra maps $\text{Alg}(B, A)$ in $\text{Hom}(C, A)$ is closed under convolution, and thus forms a (convolution) semigroup.*

DEFINITION 0.4. If H is a bialgebra and there exists a two-sided (necessarily unique) convolution inverse S for the identity map $I: H \rightarrow H$ in the algebra $\text{Hom}(H, H)$, then S is called an *antipode* of H . A bialgebra that has an antipode is a *Hopf algebra*.

PROPOSITION 0.5. *Let H be a Hopf algebra and A be a commutative algebra. Then the set of algebra maps $\text{Alg}(H, A)$ is a convolution subgroup of $\text{Hom}(H, A)$ and if $R \in \text{Alg}(H, A)$ then the convolution inverse R^{-1} of R is given by*

$$R^{-1} = R \circ S,$$

where S is the antipode of H and $\circ$ denotes functional composition.

The reader is referred to [12] for proofs and further study of bi-, co-, and Hopf algebras.

I. INCIDENCE COALGEBRAS

Let $\mathcal{P}$ be a family of locally finite partially ordered sets. We begin by considering certain equivalence relations on the collection $\text{Seg}(\mathcal{P})$ of all segments of posets in $\mathcal{P}$. It should be emphasized that two segments from different posets in $\mathcal{P}$ may be related by such an equivalence relation. We will only be considering relations under which the resulting collection of equivalence classes actually forms a set.

DEFINITION 1.1. An equivalence relation $\sim$ on $\text{Seg}(\mathcal{P})$ is *order compatible* if whenever $[x, y] \sim [u, v]$ there exists a bijection ϕ , depending in general on $[x, y]$, of $[x, y]$ onto $[u, v]$ such that for all $z \in [x, y]$,

$$[x, z] \sim [\phi(x), \phi(z)]$$

and

$$[z, y] \sim [\phi(z), \phi(y)].$$

One of the most obvious examples of an order compatible relation is isomorphism. That is, set $[x, y] \sim [u, v]$ if and only if $[x, y]$ and $[u, v]$ are isomorphic as posets. It is tempting to conjecture that isomorphism is the strongest possible order compatible relation and even perhaps that the

map ϕ above must be an isomorphism of posets. The following example indicates that this is not the case.

EXAMPLE 1.2. Take $\mathcal{P} = \{P_1, P_2\}$, where P_1 and P_2 have the Hasse diagrams depicted in Fig. 1.

Set $[x, y] \sim [u, v]$ whenever $[x, y]$ and $[u, v]$ are isomorphic or whenever $[x, y] = P_1$ and $[u, v] = P_2$. Then $\sim$ is order compatible, while P_1 and P_2 are nonisomorphic, related segments.

If $\mathcal{P}$ is a family of posets (always tacitly assumed to be locally finite) with order compatible relation $\sim$ on $\text{Seg}(\mathcal{P})$ we will call the nonempty equivalence classes of $\text{Seg}(\mathcal{P})/\sim$ *types* and the set of types will be denoted by $\mathcal{P}_\sim$. If Q is in $\text{Seg}(\mathcal{P})$ then the type, or *class*, of Q will be denoted by $\text{cl}(Q)$. Also, if $\sigma \in Q$ then we define types σ^- and σ^+ in $\mathcal{P}_\sim$ by

$$\sigma^- = \text{cl}([\hat{0}, \sigma])$$

and

$$\sigma^+ = \text{cl}([\sigma, \hat{1}]).$$

σ^- and σ^+ will be referred to as the *type below* σ and the *type above* σ , respectively.

We now define the *reduced incidence coalgebra* $C(\mathcal{P}_\sim)$ of the family $\mathcal{P}$ modulo the equivalence relation $\sim$ to be the free vector space spanned by the indeterminates X^α , where $\alpha \in \mathcal{P}_\sim$; with diagonalization Δ and counit ε given by

$$\Delta(X^\alpha) = \sum_{\sigma \in P} X^{\sigma^-} \otimes X^{\sigma^+}, \quad (1.3)$$

where $\text{cl}(P) = \alpha$, and

$$\varepsilon(X^\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ consists of one-point segments,} \\ 0 & \text{otherwise.} \end{cases} \quad (1.4)$$

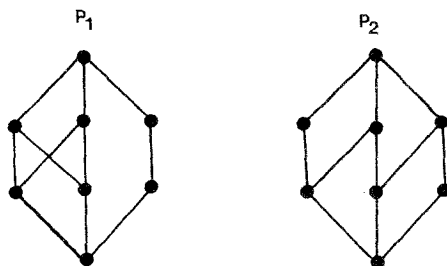


FIGURE 1

Note that Δ is well defined by (1.3) if and only if $\sim$ is an order compatible relation. ε obviously satisfies the counitary property $(I \otimes \varepsilon) \circ \Delta = (\varepsilon \otimes I) \circ \Delta = I$, where $I: C(\mathcal{P}_\sim) \rightarrow C(\mathcal{P}_\sim)$ is the identity map. To see that Δ is coassociative we note that both $(\Delta \otimes I) \circ \Delta(X^\alpha)$ and $(I \otimes \Delta) \circ \Delta(X^\alpha)$ are equal to

$$\sum_{\substack{\sigma, \tau \in P \\ \sigma \leq \tau}} X^{\sigma^-} \otimes X^{\text{cl}([\sigma, \tau])} \otimes X^{\tau^+}.$$

Hence $(C(\mathcal{P}_\sim), \Delta, \varepsilon)$ is a coalgebra.

When $\sim$ is the isomorphism relation then $C(\mathcal{P}_\sim)$ will be called the *standard reduced incidence coalgebra* of $\mathcal{P}$. If $\text{Seg}(\mathcal{P})$ is actually a set and $\sim$ is the trivial relation (i.e., every segment is only related to itself) then we write $C(\mathcal{P})$ instead of $C(\mathcal{P}_\sim)$ and this is called the *full* of *unreduced* incidence coalgebra of $\mathcal{P}$.

In the case that $\mathcal{P}$ consists of a single poset P the dual, convolution algebra $C^*(\mathcal{P})$ is precisely the incidence algebra $I(P)$ of P , and the algebra $C^*(\mathcal{P}_\sim)$ is the reduced incidence algebra $R(P, \sim)$ of P modulo $\sim$ (see Doubilet *et al.* [4]). Otherwise, the algebra $C^*(\mathcal{P}_\sim)$ gives us a simple notion of a reduced incidence algebra for a family of posets, whose construction is more direct than that of the large incidence algebras introduced in [4].

Now suppose $\mathcal{P}$ is a family of posets with full incidence coalgebra $C(\mathcal{P})$. If $\sim$ is an equivalence relation on $\text{Seg}(\mathcal{P})$ let $J_\sim$ be the linear subspace of $C(\mathcal{P})$ which is spanned by all elements of the form $X^P - X^Q$, where $P \sim Q$ in $\text{Seg}(\mathcal{P})$. The order compatible relations on $\text{Seg}(\mathcal{P})$ are characterized by the following proposition.

PROPOSITION 1.5. *The equivalence relation $\sim$ is order compatible if and only if the subspace $J_\sim$ is a coideal of $C(\mathcal{P})$, in which case the reduced incidence coalgebra $C(\mathcal{P}_\sim)$ is the quotient coalgebra $C(\mathcal{P})/J_\sim$.*

Proof. First note that $C(\mathcal{P}_\sim)$ and $C(\mathcal{P})/J_\sim$ are identical as vector spaces, and let $\pi: C(\mathcal{P}) \rightarrow C(\mathcal{P}_\sim)$ be the natural projection. If $\sim$ is order compatible then $C(\mathcal{P}_\sim)$ becomes a coalgebra with diagonalization Δ' and augmentation ε' given by

$$\Delta'(X^\alpha) = \sum_{\sigma \in P} X^{\sigma^-} \otimes X^{\sigma^+}, \quad (1.6)$$

where $\text{cl}(P) = \alpha$ and

$$\varepsilon'(X^\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ consists of one-point segments,} \\ 0 & \text{otherwise.} \end{cases}$$

This can be rewritten as $\Delta' \circ \pi = \pi \otimes \pi \circ \Delta$ and $\varepsilon' \circ \pi = \varepsilon$, in other words π is a coalgebra map. Since π has kernel $J_\sim$, it follows that $J_\sim$ is a coideal. Conversely, if $J_\sim$ is a coideal then Δ', ε' above give $C(\mathcal{P}_\sim)$ the unique coalgebra structure such that $\pi: C(\mathcal{P}) \rightarrow C(\mathcal{P}_\sim)$ is a coalgebra map. In particular, Δ' is well-defined by (1.6); therefore $\sim$ is an order compatible relation. ■

In most examples of reduced incidence coalgebras $C(\mathcal{P}_\sim)$ which are of combinatorial interest, the family $\mathcal{P}$ consists of posets P where each P is a set of combinatorial structures (e.g., subsets, partitions, graphs, etc.) on some underlying set S . Often a binary operation $\oplus$ can be defined on $\mathcal{P}$ as follows: if P and Q are posets of structures on sets S and T , respectively, then $P \oplus Q$ is the set of structures on the union $S \cup T$. Quite frequently this operation will respect the order compatible relation $\sim$ and thus yield a well-defined, associative, binary operation $+$ on the set of types $\mathcal{P}_\sim$. This motivates

DEFINITION 1.7. If the set of types $\mathcal{P}_\sim$ is a semigroup with binary operation $+$, then the *induced algebra structure on $C(\mathcal{P}_\sim)$* is given by setting

$$X^\alpha X^\beta = X^{\alpha + \beta} \quad (1.8)$$

for α, β in $\mathcal{P}_\sim$.

We now describe exactly when a semigroup of types $\mathcal{P}_\sim$ with order compatible relation $\sim$ is such that the reduced incidence coalgebra $C(\mathcal{P})$ becomes a bialgebra with the induced algebra structure.

PROPOSITION 1.9. Let $\mathcal{P}$ be a family of posets with order compatible relation $\sim$; and suppose the set of types $\mathcal{P}_\sim$ is a semigroup with binary operation $+$. Then $C(\mathcal{P}_\sim)$ becomes a bialgebra with the induced algebra structure if and only if whenever $\alpha = \alpha_1 + \alpha_2$ in $\mathcal{P}_\sim$, and we choose representatives $P \in \alpha$, $P_1 \in \alpha_1$, and $P_2 \in \alpha_2$ then there exists a bijection ϕ , depending in general on P_1 and P_2 , from the set $P_1 \times P_2$ onto P such that for all (σ_1, σ_2) in $P_1 \times P_2$,

$$\phi(\sigma_1, \sigma_2)^- = \sigma_1^- + \sigma_2^-,$$

and

$$\phi(\sigma_1, \sigma_2)^+ = \sigma_1^+ + \sigma_2^+.$$

Proof. Suppose the hypotheses are true. We must then show that

$\Delta(X^{\alpha_1} X^{\alpha_2}) = \Delta(X^{\alpha_1}) \Delta(X^{\alpha_2})$ and $\varepsilon(X^{\alpha_1} X^{\alpha_2}) = \varepsilon(X^{\alpha_1}) \varepsilon(X^{\alpha_2})$ for all α_1, α_2 in $\mathcal{P}_\sim$. If $\alpha = \alpha_1 + \alpha_2$ and we choose P of type α then

$$\begin{aligned} \Delta(X^{\alpha_1} X^{\alpha_2}) &= \Delta(X^\alpha) \\ &= \sum_{\sigma \in P} X^{\sigma^-} \otimes X^{\sigma^+}. \end{aligned}$$

Now choose representatives $P_1 \in \alpha_1$ and $P_2 \in \alpha_2$ and let $\phi: P_1 \times P_2 \rightarrow P$ be a bijection satisfying (1.10). The above sum can then be written

$$\begin{aligned} &\sum_{\sigma_1 \in P_1} \sum_{\sigma_2 \in P_2} X^{\phi(\sigma_1, \sigma_2)^-} \otimes X^{\phi(\sigma_1, \sigma_2)^+} \\ &= \sum_{\sigma_1 \in P_1} \sum_{\sigma_2 \in P_2} X^{\sigma_1^- + \sigma_2^-} \otimes X^{\sigma_1^+ + \sigma_2^+} \\ &= \left(\sum_{\sigma_1 \in P_1} X^{\sigma_1^-} \otimes X^{\sigma_1^+} \right) \left(\sum_{\sigma_2 \in P_2} X^{\sigma_2^-} \otimes X^{\sigma_2^+} \right) \\ &= \Delta(X^{\alpha_1}) \Delta(X^{\alpha_2}). \end{aligned}$$

The identity $\varepsilon(X^{\alpha_1} X^{\alpha_2}) = \varepsilon(X^{\alpha_1}) \varepsilon(X^{\alpha_2})$ is trivial. Conversely, suppose $C(\mathcal{P}_\sim)$ is a bialgebra with the induced algebra structure. Then $\Delta(X^\alpha) = \Delta(X^{\alpha_1}) \Delta(X^{\alpha_2})$ for $\alpha = \alpha_1 + \alpha_2$ and if we choose P, P_1 , and P_2 of types α, α_1 , and α_2 , respectively, this implies

$$\sum_{\sigma_1 \in P_1} \sum_{\sigma_2 \in P_2} X^{\sigma_1^- + \sigma_2^-} \otimes X^{\sigma_1^+ + \sigma_2^+} = \sum_{\sigma \in P} X^{\sigma^-} \otimes X^{\sigma^+}.$$

The equality of these two sums of basis elements of $C(\mathcal{P}_\sim) \otimes C(\mathcal{P}_\sim)$ can only hold if a bijection $\phi: P_1 \times P_2 \rightarrow P$ satisfying (1.10) exists. ■

When a semigroup of types $\mathcal{P}_\sim$ satisfies the hypotheses of Proposition 1.9, we will say that $\mathcal{P}_\sim$ has an *order compatible* semigroup structure.

EXAMPLE 1.11 (The Fàa di Bruno bialgebra). Let $\Pi(S)$ denote the set of partitions of a finite set S . If we order the elements of $\Pi(S)$ by refinement (i.e., $\pi < \tau$ in $\Pi(S)$ if and only if every block of π is contained in a block of τ) then $\Pi(S)$ becomes a lattice. To each interval $[\pi, \tau]$ in $\Pi(S)$ we associate a vector $\alpha = (\alpha_1, \alpha_2, \dots)$, where α_i is equal to the number of blocks of τ which are unions of exactly i blocks of π .

Now let $\mathcal{P}$ be the family of all lattices of partitions of finite sets. We define an equivalence relation $\sim$ on $\text{Seg}(\mathcal{P})$ by setting $[\sigma, \lambda] \sim [\pi, \tau]$ if and only if the associated vectors of these two segments are the same. This is an order compatible equivalence relation on $\mathcal{P}$. The types in $\mathcal{P}_\sim$ are vec-

tors $\alpha = (\alpha_1, \alpha_2, \dots)$ of non-negative integers with finitely many non-zero entries. There is an (associative) addition of types given by

$$\alpha + \beta = (\alpha_1 + \beta_1, \alpha_2 + \beta_2, \dots)$$

for $\alpha = (\alpha_1, \alpha_2, \dots)$ and $\beta = (\beta_1, \beta_2, \dots)$ in $\mathcal{P}_\sim$. Now suppose $\alpha = \alpha_1 + \alpha_2$ in $\mathcal{P}_\sim$. Choose representatives $[\pi, \tau] \in \alpha$, $[\pi_1, \tau_1] \in \alpha_1$, and $[\pi_2, \tau_2] \in \alpha_2$. Then $\tau_1 \cup \tau_2$ partitions the set of blocks $\pi_1 \cup \pi_2$, and since $\alpha = \alpha_1 + \alpha_2$, we can choose a bijection ϕ from the set $\pi_1 \cup \pi_2$ onto the set π which maps blocks of $\tau_1 \cup \tau_2$ onto blocks of τ . ϕ naturally induces a mapping $\bar{\phi}$ from $[\pi_1, \tau_1] \times [\pi_2, \tau_2]$ onto $[\pi, \tau]$ which satisfies the hypotheses of Proposition 1.9. So if we define multiplication in the reduced incidence coalgebra $C(\mathcal{P}_\sim)$, by $X^\alpha X^\beta = X^{\alpha+\beta}$ then $C(\mathcal{P}_\sim)$ becomes a bialgebra.

This bialgebra, which will be denoted by $\mathcal{F}$, is known as the *Faà di Bruno bialgebra*.

Finally we remark that if we set $X^\alpha = X_1^{\alpha_1} X_2^{\alpha_2} X_3^{\alpha_3} \cdots$ for $\alpha \in \mathcal{P}_\sim$, then as an algebra $\mathcal{F}$ is isomorphic to $K[x_1, x_2, \dots]$, the algebra of polynomials in a countably infinite number of variables.

A convenient notational device which gives us another, sometimes more manageable, way of expressing the diagonalization (1.3) is that of *section coefficients*. Given a family $\mathcal{P}$ of posets with order compatible relation $\sim$, and types α, β, γ in $\mathcal{P}_\sim$, we define the section coefficient $(\alpha | \beta, \gamma)$ to be the number of elements σ in a representative P of α such that $\sigma^- = \beta$ and $\sigma^+ = \gamma$. Note that $\sim$ is an order compatible relation on $\text{Seg}(\mathcal{P})$ if and only if the section coefficients $(\alpha | \beta, \gamma)$ are well defined for all types α, β, γ in $\mathcal{P}_\sim$. The diagonalization can now be written

$$\Delta(X^\alpha) = \sum_{\beta, \gamma} (\alpha | \beta, \gamma) X^\beta \otimes X^\gamma,$$

where the sum ranges over all β and γ in $\mathcal{P}_\sim$. The fact that this is a finite sum and that Δ is coassociative implies the following properties of section coefficients.

PROPOSITION 1.12. (i) $(\alpha | \beta, \gamma) = 0$ for all but finitely many types $\beta, \gamma \in \mathcal{P}_\sim$. (ii) The section coefficients satisfy

$$\sum_{\rho} (\alpha | \beta, \rho)(\rho | \gamma, \lambda) = \sum_{\eta} (\alpha | \eta, \lambda)(\eta | \beta, \gamma) \quad (1.13)$$

for all types $\alpha, \beta, \gamma, \lambda$ in $\mathcal{P}_\sim$.

Proof. (i) is trivial since all segments of any type in $\mathcal{P}_\sim$ are finite. For (ii) we calculate

$$(I \otimes \Delta) \circ \Delta(X^\alpha) = \sum_{\beta, \rho} \sum_{\gamma, \lambda} (\alpha | \beta, \rho)(\rho | \gamma, \lambda) X^\beta \otimes X^\gamma \otimes X^\lambda$$

and

$$(\Delta \otimes I) \circ \Delta(X^\alpha) = \sum_{\eta, \lambda} \sum_{\beta, \gamma} (\alpha | \eta, \lambda)(\eta | \beta, \gamma) X^\beta \otimes X^\gamma \otimes X^\lambda.$$

These two sums are equal by coassociativity, and (1.13) follows by equating coefficients of basis elements.

The common value of the sums in (1.13) is called a *multisection coefficient* and denoted by $(\alpha | \beta, \gamma, \lambda)$. We can define higher order multisection coefficients $(\alpha | \beta_1, \dots, \beta_n)$ for $\alpha, \beta_1, \dots, \beta_n$ in $\mathcal{P}_\sim$ to be the number of chains $\hat{0} = \sigma_0 < \sigma_1 < \dots < \sigma_n = \hat{1}$ in a segment of type α such that $\text{cl}([\sigma_{i-1}, \sigma_i]) = \beta_i$ for $1 \leq i \leq n$.

PROPOSITION 1.14. *If the family of types $\mathcal{P}_\sim$ is a semigroup with binary operation $+$ then $C(\mathcal{P}_\sim)$ is a bialgebra with the induced multiplication if and only if the section coefficients satisfy*

$$(\alpha_1 + \alpha_2 | \beta, \gamma) = \sum_{\substack{\beta_1 + \beta_2 = \beta \\ \gamma_1 + \gamma_2 = \gamma}} (\alpha_1 | \beta_1, \gamma_1)(\alpha_2 | \beta_2, \gamma_2) \quad (1.15)$$

for all $\alpha_1, \alpha_2, \beta, \gamma$ in $\mathcal{P}_\sim$. (Such section coefficients are known as bisection coefficients.)

Proof. We calculate

$$\Delta(X^{\alpha_1} X^{\alpha_2}) = \Delta(X^{\alpha_1 + \alpha_2}) = \sum_{\beta, \gamma} (\alpha_1 + \alpha_2 | \beta, \gamma) X^\beta \otimes X^\gamma$$

while

$$\begin{aligned} \Delta(X^{\alpha_1}) \Delta(X^{\alpha_2}) &= \left(\sum_{\beta_1, \gamma_1} (\alpha_1 | \beta_1, \gamma_1) X^{\beta_1} \otimes X^{\gamma_1} \right) \\ &\quad \times \left(\sum_{\beta_2, \gamma_2} (\alpha_2 | \beta_2, \gamma_2) X^{\beta_2} \otimes X^{\gamma_2} \right) \\ &= \sum_{\beta_1, \gamma_1} \sum_{\beta_2, \gamma_2} (\alpha_1 | \beta_1, \gamma_1)(\alpha_2 | \beta_2, \gamma_2) X^{\beta_1 + \beta_2} \otimes X^{\gamma_1 + \gamma_2}, \end{aligned}$$

hence $\Delta(X^{\alpha_1} X^{\alpha_2}) = \Delta(X^{\alpha_1}) \Delta(X^{\alpha_2})$ if and only if (1.15) is true.

EXAMPLE 1.16 (Binomial bialgebra). The binomial bialgebra $\mathcal{B}$ is the standard reduced incidence coalgebra for the family of lattices of subsets of finite sets. Types correspond to non-negative integers by cardinality of the underlying set of a representative. Addition of types is simply addition of integers. It is trivial to check that the hypotheses of Proposition 1.9 are

satisfied and thus $\mathcal{B}$ is indeed a bialgebra. The section coefficients are given by

$$(n \mid j, k) = \begin{cases} \binom{n}{j} = \binom{n}{k} & \text{if } j + k = n, \\ 0 & \text{otherwise.} \end{cases}$$

The multisection coefficients are

$$(n, j_1, \dots, j_k) = \begin{cases} \binom{n}{j_1, \dots, j_k} & \text{if } j_1 + \dots + j_k = n, \\ 0 & \text{otherwise.} \end{cases}$$

Because $\mathcal{B}$ is a bialgebra, Proposition 1.14 implies

$$(n + m \mid j, k) = \sum_{j_1 + j_2 = j} \sum_{k_1 + k_2 = k} (n \mid j_1, k_1)(m \mid j_2, k_2),$$

so in particular when $j + k = n + m$ we have

$$\binom{n + m}{j} = \sum_{j_1 + j_2 = j} \binom{n}{j_1} \binom{m}{j_2},$$

which is the well-known Vandermonde convolution identity. Finally we remark that the diagonalization in $\mathcal{B}$ is given by

$$\Delta X^n = \sum_{k=0}^n \binom{n}{k} X^k \otimes X^{n-k}.$$

II. HEREDITARY BIALGEBRAS

In this section we give a simple description of those families of posets that have standard reduced incidence algebras which are in fact bialgebras.

DEFINITION 2.1. Let $\mathcal{P}$ be a family of posets and let $\sim$ be the isomorphism relation on $\text{Seg}(\mathcal{P})$. Then $\mathcal{P}$ is a *hereditary class* of posets if whenever α_1 and α_2 are types in $\mathcal{P}_\sim$, $\text{cl}(P_1) = \alpha_1$, and $\text{cl}(P_2) = \alpha_2$ then $\text{cl}(P_1 \times P_2)$ is in $\mathcal{P}_\sim$ where $P_1 \times P_2$ is the direct product of the segments P_1 and P_2 .

In other words, if we define the sum of two isomorphism classes of segments to be the isomorphism class of the direct product of two representatives, then $\mathcal{P}$ is a hereditary class if $\mathcal{P}_\sim$ is closed under sums, where $\sim$ is the isomorphism relation. We now have the following general result:

PROPOSITION 2.2. *If $\mathcal{P}$ is any hereditary class of posets than the standard reduced incidence coalgebra $C(\mathcal{P}_\sim)$ is a bialgebra under the induced multiplication. Such bialgebras will be called hereditary bialgebras.*

Proof. Let $\alpha = \alpha_1 + \alpha_2$ in $\mathcal{P}_\sim$. Choose representatives P, P_1 , and P_2 of α, α_1 , and α_2 , respectively. By the definition of $\alpha_1 + \alpha_2$ in $\mathcal{P}_\sim$, there must exist an isomorphism ϕ from the direct product $P_1 \times P_2$ onto P . But then for $(\sigma_1, \sigma_2) \in P_1 \times P_2$ we have

$$\phi(\sigma_1, \sigma_2)^+ = \sigma_1^+ + \sigma_2^+$$

and

$$\phi(\sigma_1, \sigma_2)^- = \sigma_1^- + \sigma_2^-,$$

by the definition of $+$, and the fact that ϕ is an isomorphism. Thus by Proposition 1.9, $C(\mathcal{P}_\sim)$ is a bialgebra.

II. FREE INCIDENCE BIALGEBRAS

Let $\mathcal{P}$ be a family of posets with order compatible relation $\sim$. Define $\mathcal{P}'$ to be the family of all finite products $P_1 \times \cdots \times P_k$, where the P_i are posets in $\mathcal{P}$. Note that $\text{Seg}(\mathcal{P}')$ then consists of all finite products $Q_1 \times \cdots \times Q_k$, where the Q_i are in $\text{Seg}(\mathcal{P})$. We define an equivalence relation $\sim'$ on $\text{Seg}(\mathcal{P}')$ by setting

$$Q_1 \times \cdots \times Q_k \sim' Q'_1 \times \cdots \times Q'_m,$$

whenever $k = m$ and there exists a permutation π of $\{1, \dots, k\}$ such that

$$Q_i \sim Q'_{\pi(i)} \quad \text{for } 1 \leq i \leq k.$$

PROPOSITION 3.1. *The equivalence relation $\sim'$ on $\text{Seg}(\mathcal{P}')$ is order compatible.*

Proof. The proof is immediate from the definition of $\sim'$ and Definition 1.1. ■

If $Q = Q_1 \times \cdots \times Q_k \in \text{Seg}(\mathcal{P}')$, where $\text{cl}(Q_i) = \alpha_i \in \mathcal{P}_\sim$ for $1 \leq i \leq k$, then we write the basis element $X^{\text{cl}(Q)}$ of the reduced incidence coalgebra $C(\mathcal{P}'_\sim)$ as the monomial $X_{\alpha_1} X_{\alpha_2} \cdots X_{\alpha_k}$. It is clear that this defines a one-to-one correspondence between types in $\mathcal{P}'_\sim$ and the set of monomials in the variables $X_\alpha, \alpha \in \mathcal{P}_\sim$. If $Q = Q_1 \times \cdots \times Q_k$ and $Q' = Q'_1 \times \cdots \times Q'_m$ are in $\text{Seg}(\mathcal{P}')$ define $\text{cl}(Q) + \text{cl}(Q')$ in $\mathcal{P}'_\sim$ by

$$\text{cl}(Q) + \text{cl}(Q') = \text{cl}(Q_1 \times \cdots \times Q_k \times Q'_1 \times \cdots \times Q'_m). \quad (3.2)$$

Thus we have a well-defined commutative and associative addition of types in $\mathcal{P}'_{\sim}$. Note that the semigroup $\mathcal{P}'_{\sim}$ has an identity element. Namely the class of the empty product of segments in $\mathcal{P}'$ which, by convention, consists of a single, one point, representative.

PROPOSITION 3.3. *The reduced incidence coalgebra $C(\mathcal{P}'_{\sim})$ is a bialgebra with the induced algebra structure.*

Proof. Let $X_{\alpha_1} \cdots X_{\alpha_k}$ be a basis element of $C(\mathcal{P}'_{\sim})$ and let $Q = Q_1 \times \cdots \times Q_k$ where $\text{cl}(Q_i) = \alpha_i$ in $\mathcal{P}_{\sim}$, for $1 \leq i \leq k$. Then the diagonalization Δ' in $C(\mathcal{P}'_{\sim})$ is given by

$$\begin{aligned} \Delta'(X_{\alpha_1} \cdots X_{\alpha_k}) &= \sum_{\sigma \in Q} X^{\sigma^-} \otimes X^{\sigma^+} \\ &= \sum_{(\sigma_1, \dots, \sigma_k) \in Q_1 \times \cdots \times Q_k} X_{\sigma_1^-} \cdots X_{\sigma_k^-} \otimes X_{\sigma_1^+} \cdots X_{\sigma_k^+} \\ &= \left(\sum_{\sigma_1 \in Q_1} X_{\sigma_1^-} \otimes X_{\sigma_1^+} \right) \cdots \left(\sum_{\sigma_k \in Q_k} X_{\sigma_k^-} \otimes X_{\sigma_k^+} \right) \\ &= \Delta'(X_{\alpha_1}) \cdots \Delta'(X_{\alpha_k}). \end{aligned}$$

For the counit ε' we have

$$\begin{aligned} \varepsilon'(X_{\alpha_1} \cdots X_{\alpha_k}) &= \begin{cases} 1 & \text{if } |Q| = 1 \\ 0 & \text{otherwise} \end{cases} \\ &= \begin{cases} 1 & \text{if } |Q_1| = \cdots = |Q_k| = 1 \\ 0 & \text{otherwise} \end{cases} \\ &= \varepsilon'(X_{\alpha_1}) \cdots \varepsilon'(X_{\alpha_k}). \end{aligned}$$

Hence Δ' and ε' are algebra maps. ■

From now on the bialgebra $C(\mathcal{P}'_{\sim})$ will be denoted by $B(\mathcal{P}_{\sim})$ and will be called the *free incidence bialgebra of the family $\mathcal{P}$ modulo $\sim$* . The inclusion $\text{Seg}(\mathcal{P}) \subseteq \text{Seg}(\mathcal{P}')$ induces an injective mapping $\iota: C(\mathcal{P}_{\sim}) \rightarrow B(\mathcal{P}_{\sim})$, where

$$\iota: X^{\alpha} \rightarrow X_{\alpha} \quad \text{for } \alpha \in \mathcal{P}_{\sim}.$$

PROPOSITION 3.4. *The injection $\iota: C(\mathcal{P}_{\sim}) \rightarrow B(\mathcal{P}_{\sim})$ is a coalgebra map.*

Proof. Suppose $Q \in \text{Seg}(\mathcal{P})$ and $\text{cl}(Q) = \alpha$. Then

$$\begin{aligned} \Delta' \circ \iota(X^\alpha) &= \Delta'(X_\alpha) \\ &= \sum_{\sigma \in Q} X_{\sigma^-} \otimes X_{\sigma^+} \\ &= (\iota \otimes \iota) \sum_{\sigma \in Q} X^{\sigma^-} \otimes X^{\sigma^+} \\ &= (\iota \otimes \iota) \circ \Delta X^\alpha. \end{aligned}$$

So $\Delta' \circ \iota = (\iota \otimes \iota) \circ \Delta$. The identity $\varepsilon' \circ \iota = \varepsilon$ is trivial, hence ι is a coalgebra map. ■

It is evident from the construction of $B(\mathcal{P}_\sim)$ that the mapping $\iota: C(\mathcal{P}_\sim) \rightarrow B(\mathcal{P}_\sim)$ satisfies the following universal property:

PROPOSITION 3.5. *Given any linear map f from $C(\mathcal{P}_\sim)$ into a commutative algebra A with unit, there exists a unique algebra map $\bar{f}: B(\mathcal{P}_\sim) \rightarrow A$ such that the diagram*

$$\begin{array}{ccc} C(\mathcal{P}_\sim) & \xrightarrow{\iota} & B(\mathcal{P}_\sim) \\ & \searrow f & \swarrow \bar{f} \\ & A & \end{array}$$

commutes.

Recall that if B is a bialgebra then the set of elements in the dual algebra B^* which are algebra maps is closed under convolution.

DEFINITION 3.6. If B is a bialgebra then the *semigroup of multiplicative functions on B* , denoted by $M(B)$, is the set of elements in B^* which are algebra maps, which convolution as binary operation.

In the special case when the commutative algebra A is the scalar field K , Proposition 3.5 gives a bijective correspondence $f \rightarrow \bar{f}$ between the dual algebra $C^*(\mathcal{P}_\sim)$ and the semigroup of multiplicative functions $M(B(\mathcal{P}_\sim))$. This correspondence is in fact much stronger than a bijection.

PROPOSITION 3.7. *The bijection $f \rightarrow \bar{f}$ from $C^*(\mathcal{P}_\sim)$ to $M(B(\mathcal{P}_\sim))$ is an isomorphism of (convolution) semigroups.*

Proof. We need to show for g, f in $C^*(\mathcal{P}_\sim)$ that $\overline{g * f} = \bar{g} * \bar{f}$. Since $\overline{g * f}$ is determined uniquely by the condition $\overline{g * f} \circ \iota = g * f$, it is therefore sufficient to show that $\bar{g} * \bar{f} \circ \iota = g * f$. If we let $\mu_K: K \otimes K \rightarrow K$ denote mul-

multiplication in the base field K , then by definition of convolution in $B^*(\mathcal{P}_\sim)$ we have

$$\begin{aligned}\bar{g} * \bar{f} \circ \iota &= \mu_K \circ (\bar{g} \otimes \bar{f}) \circ \Delta' \circ \iota \\ &= \mu_K \circ (\bar{g} \otimes \bar{f}) \circ (\iota \otimes \iota) \circ \Delta,\end{aligned}$$

because ι is a coalgebra map. But this is equal to

$$\mu_K \circ (g \otimes f) \circ \Delta,$$

which is the definition of $g * f$ in $C^*(\mathcal{P}_\sim)$. Thus the map $f \rightarrow \bar{f}$ is an isomorphism.

IV. THE RECURSIVE FORMULA FOR THE ANTIPODE OF AN INCIDENCE COALGEBRA

Let $\mathcal{P}$ be a family of posets with order compatible relation $\sim$, and suppose that the set of types $\mathcal{P}_\sim$ has an order compatible semigroup structure (so that $C(\mathcal{P}_\sim)$ becomes a bialgebra when given the induced algebra structure). We now give a necessary and sufficient condition on the semigroup $\mathcal{P}_\sim$ such that the bialgebra $C(\mathcal{P}_\sim)$ actually becomes a Hopf algebra. More importantly, however, in this case we give an explicit recursive formula for the antipode S of the Hopf algebra $C(\mathcal{P}_\sim)$.

THEOREM 4.1 (Recursive formula for the antipode.) *The bialgebra $C(\mathcal{P}_\sim)$ is a Hopf algebra if and only if the semigroup $\mathcal{P}_\sim$ has an identity element and all the one-point types in $\mathcal{P}_\sim$ (i.e., types which consist of one-point segments) have two-sided inverses in $\mathcal{P}_\sim$. In which case, the antipode S of $C(\mathcal{P}_\sim)$ is given recursively as follows:*

If α is a one-point type then

$$S(X^\alpha) = X^{-\alpha}, \quad (4.2)$$

where $-\alpha$ is the inverse of α in $\mathcal{P}_\sim$.

Otherwise, we pick a representative P of α and

$$S(X^\alpha) = -X^{-\hat{0}^-} \sum_{\substack{\sigma \in P \\ \sigma > \hat{0}}} X^{\sigma^-} S(X^{\sigma^+}) \quad (4.3)$$

(where $-\hat{0}^-$ is the additive inverse of the one-point type $\hat{0}^-$). Also we have the dual form

$$S(X^\alpha) = - \left(\sum_{\substack{\sigma \in P \\ \sigma < \hat{1}}} S(X^{\sigma^-}) X^{\sigma^+} \right) X^{-\hat{1}^+}. \quad (4.4)$$

Proof. First we remark that (4.3) and (4.4) are both proper recursion formulas, because on the right-hand sides of these equations we are evaluating S only on basis elements X^β , where β consists of segments of cardinality strictly less than that of P . Now the antipode $S: C(\mathcal{P}_\sim) \rightarrow C(\mathcal{P}_\sim)$ is uniquely defined by the conditions

$$S * I(X^\alpha) = \varepsilon(X^\alpha) e \quad (4.5)$$

and

$$I * S(X^\alpha) = \varepsilon(X^\alpha) e, \quad (4.6)$$

where $I: C(\mathcal{P}_\sim) \rightarrow C(\mathcal{P}_\sim)$ is the identity map and e is the (multiplicative) identity element of $C(\mathcal{P}_\sim)$.

If α is a one-point type then according to (4.2) $S * I(X^\alpha) = X^{-\alpha} X^\alpha = e$ and similarly $I * S(X^\alpha) = e$. Since $\varepsilon(X^\alpha) = 1$, S thus satisfies (4.5) and (4.6) for one point types α . If α is not a one-point type then $\varepsilon(X^\alpha) = 0$ and (4.5) becomes

$$S * I(X^\alpha) = 0.$$

By the definition of convolution in $\text{Hom}(C(\mathcal{P}_\sim), C(\mathcal{P}_\sim))$ this can be written as

$$\sum_{\sigma \in P} S(X^{\sigma^-}) X^{\sigma^+} = 0$$

or

$$S(X^\alpha) X^{\hat{1}^+} = - \sum_{\substack{\sigma \in P \\ \sigma < \hat{1}}} S(X^{\sigma^-}) X^{\sigma^+}$$

from which we obtain (4.4) by multiplying both sides by $X^{-\hat{1}^+}$ on the right. Similarly we see that (4.3) is equivalent to Eq. (4.6). Thus we have shown that (4.3) and (4.4) define right- and left-convolution inverses for I in $\text{Hom}(C(\mathcal{P}_\sim), C(\mathcal{P}_\sim))$, respectively. Since convolution is an associative operation, right and left inverses must be equal; hence (4.3) and (4.4) define the same element $S \in \text{Hom}(C(\mathcal{P}_\sim), C(\mathcal{P}_\sim))$, which is the unique two-sided convolution inverse of I .

Conversely, the necessity of one-point types having two-sided inverses in $\mathcal{P}_\sim$ follows from the conditions $S * I(X^\alpha) = S(X^\alpha) X^\alpha = e$ and $I * S(X^\alpha) = X^\alpha S(X^\alpha) = e$ for all one-point types α . ■

As an immediate consequence of Theorem 4.1, we have the following important class of Hopf algebras.

COROLLARY 4.7. *If $\mathcal{P}$ is a hereditary class of posets then the standard reduced incidence coalgebra $C(\mathcal{P}_\sim)$ is a Hopf algebra. Such Hopf algebras will be called hereditary Hopf algebras.*

Proof. We know from Proposition 2.2 that $C(\mathcal{P}_\sim)$ is a bialgebra. The only one-point type in $\mathcal{P}_\sim$ is the isomorphism class of a one-point poset; and this type is the identity element of the semigroup $\mathcal{P}_\sim$. Thus Theorem 4.1 applies.

V. INCIDENCE HOPF ALGEBRAS

Suppose $\mathcal{P}$ is a family of posets with order compatible relation $\sim$, and the set of types $\mathcal{P}_\sim$ has an order compatible semigroup structure, so that the reduced incidence coalgebra $C(\mathcal{P}_\sim)$ is a bialgebra with the induced multiplication. In this section we consider the problem of embedding $C(\mathcal{P}_\sim)$ in a Hopf algebra. We show that under certain conditions on the semigroup $\mathcal{P}_\sim$, we can actually imbed $\mathcal{P}_\sim$ in a semigroup $\tilde{\mathcal{P}}_\sim$ of types of posets such that the reduced incidence coalgebra $C(\tilde{\mathcal{P}}_\sim)$ is a Hopf algebra with the induced multiplication, and the embedding of $\mathcal{P}_\sim$ into $\tilde{\mathcal{P}}_\sim$ lifts to an injective bialgebra map ι from $C(\mathcal{P}_\sim)$ into $C(\tilde{\mathcal{P}}_\sim)$.

Now suppose the set of types $\mathcal{P}_\sim$ has a commutative order compatible semigroup structure. Also suppose that the one point types of $\mathcal{P}_\sim$ are *cancellable*, that is

$$\alpha + \gamma = \beta + \gamma \Rightarrow \alpha = \beta \quad (5.1)$$

whenever $\alpha, \beta \in \mathcal{P}_\sim$ and γ is a one-point type in $\mathcal{P}_\sim$. If the semigroup $\mathcal{P}_\sim$ does not have an identity element we add a one-point poset to $\mathcal{P}$ and declare its class to be the identity 0 in $\mathcal{P}_\sim$. Let S be the subset of $\mathcal{P}_\sim$ consisting of all one-point types that have no inverse in $\mathcal{P}_\sim$, and the identity 0 of $\mathcal{P}_\sim$. $\tilde{\mathcal{P}}$ is then defined to be the family of all products $P \times Q$ where $P \in \mathcal{P}$, $Q \in \text{Seg}(\mathcal{P})$ and $\text{cl}(Q)$ is in S . We define a relation $\approx$ on $\text{Seg}(\tilde{\mathcal{P}})$ by setting

$$P_1 \times Q_1 \approx P_2 \times Q_2 \quad (5.2)$$

if and only if

$$\text{cl}(P_1) + \text{cl}(Q_2) = \text{cl}(P_2) + \text{cl}(Q_1) \quad \text{in } \mathcal{P}_\sim.$$

It is easy to show that $\approx$ is a transitive relation by using the commutativity of $\mathcal{P}_\sim$ and the cancellation rule (5.1) for types in S . Symmetry and reflexivity of $\approx$ are immediate. Thus $\approx$ is an equivalence relation on $\text{Seg}(\tilde{\mathcal{P}})$.

PROPOSITION 5.3. *The equivalence relation $\approx$ is order compatible.*

Proof. Suppose $P_1 \times Q_1 \approx P_2 \times Q_2$, where $Q_1 = \{a\}$ and $Q_2 = \{b\}$. Any bijection $\Phi: P_1 \times Q_1 \rightarrow P_2 \times Q_2$ necessarily has the form $\Phi(\sigma, a) = (\psi(\sigma), b)$, where $\psi: P_1 \rightarrow P_2$ is a bijection. So to prove $\approx$ is order compatible we must find a bijection $\psi: P_1 \rightarrow P_2$ such that for all $\sigma \in P_1$, $(\sigma, a)^+ = (\psi(\sigma), b)^+$ and $(\sigma, a)^- = (\psi(\sigma), b)^-$ in $\mathcal{P}_\approx$. In other words, we must have

$$\psi(\sigma)^+ + a^+ = \sigma^+ + b^+ \quad (5.4)$$

and

$$\psi(\sigma)^- + a^- = \sigma^- + b^-$$

in $\mathcal{P}_\approx$. (Note that $a^+ = a^- = \text{cl}(Q_1)$ and $b^+ = b^- = \text{cl}(Q_2)$ in $\mathcal{P}_\approx$.)

Let R_1 represent $\text{cl}(P_1) + \text{cl}(Q_2)$ and R_2 represent $\text{cl}(P_2) + \text{cl}(Q_1)$ in $\mathcal{P}_\approx$. Since $\mathcal{P}_\approx$ is an order compatible semigroup we have bijections $\phi_1: P_1 \times Q_2 \rightarrow R_1$ and $\phi_2: P_2 \times Q_1 \rightarrow R_2$ such that

$$\phi_1(\sigma, b)^+ = \sigma^+ + b^+,$$

$$\phi_1(\sigma, b)^- = \sigma^- + b^-$$

and

$$\phi_2(\tau, a)^+ = \tau^+ + a^+,$$

$$\phi_2(\tau, a)^- = \tau^- + a^-$$

in $\mathcal{P}_\approx$ for all $\sigma \in P_1$, $\tau \in P_2$.

Now by hypothesis we have $\text{cl}(R_1) = \text{cl}(R_2)$ in $\mathcal{P}_\approx$. So there exists a bijection $\varphi: R_1 \rightarrow R_2$ such that $\varphi(\sigma)^+ = \sigma^+$ and $\varphi(\sigma)^- = \sigma^-$ in $\mathcal{P}_\approx$ for all σ in R_1 .

Now the composition $\phi_2^{-1} \circ \varphi \circ \phi_1$ maps $P_1 \times Q_2$ onto $P_2 \times Q_1$ so we can define a bijection $\psi: P_1 \rightarrow P_2$ by

$$(\psi(\sigma), a) = \phi_2^{-1} \circ \varphi \circ \phi_1(\sigma, b)$$

for $\sigma \in P_1$. Then ψ has the desired property (5.4). ■

The semigroup structure on $\mathcal{P}_\approx$ is now defined by setting

$$\text{cl}(P_1 \times Q_1) + \text{cl}(P_2 \times Q_2) = \text{cl}(P \times Q) \quad (5.5)$$

when $\text{cl}(P) = \text{cl}(P_1) + \text{cl}(P_2)$ and $\text{cl}(Q) = \text{cl}(Q_1) + \text{cl}(Q_2)$ in $\mathcal{P}_\approx$.

PROPOSITION 5.6. *The addition of types (5.5) gives $\mathcal{P}_\approx$ a commutative, order compatible semigroup structure.*

Proof. It is trivial to check that (5.5) gives a well-defined commutative associative binary operation on the set of types $\bar{\mathcal{P}}_{\sim}$. Now suppose $\alpha = \alpha_1 + \alpha_2$ in $\bar{\mathcal{P}}_{\sim}$, if $P \times Q$, $P_1 \times Q_1$, and $P_2 \times Q_2$ represent α , α_1 , and α_2 , respectively, where $Q_1 = \{a\}$, $Q_2 = \{b\}$, and $Q = \{c\}$, we need to exhibit a bijection $\phi: (P_1 \times Q_1) \times (P_2 \times Q_2) \rightarrow P \times Q$ such that

$$\phi((\sigma_1, a), (\sigma_2, b))^+ = (\sigma_1, a)^+ + (\sigma_2, b)^+$$

and (5.7)

$$\phi((\sigma_1, a), (\sigma_2, b))^- = (\sigma_1, a)^- + (\sigma_2, b)^-$$

in $\bar{\mathcal{P}}_{\sim}$, for all $\sigma_1 \in P_1$ and $\sigma_2 \in P_2$. Since $\mathcal{P}_{\sim}$ is an order compatible semigroup and $\text{cl}(P) = \text{cl}(P_1) + \text{cl}(P_2)$, we must have a bijection $\varphi: P_1 \times P_2 \rightarrow P$ with $\varphi(\sigma_1, \sigma_2)^+ = \sigma_1^+ + \sigma_2^+$ and $\varphi(\sigma_1, \sigma_2)^- = \sigma_1^- + \sigma_2^-$ in $\mathcal{P}_{\sim}$, for all $(\sigma_1, \sigma_2) \in P_1 \times P_2$. Now define $\phi: (P_1 \times Q_1) \times (P_2 \times Q_2) \rightarrow P \times Q$ by

$$\phi((\sigma_1, a), (\sigma_2, b)) = (\varphi(\sigma_1, \sigma_2), c)$$

for all $\sigma_1 \in P_1$, $\sigma_2 \in P_2$. Then ϕ satisfies (5.7). ■

PROPOSITION 5.8. *The set of one-point types G in $\bar{\mathcal{P}}_{\sim}$ is a group under the addition $+$ of types in $\bar{\mathcal{P}}_{\sim}$. The identity element $\bar{0}$ of G is $\text{cl}(Q \times Q)$, where $Q \in S$.*

Proof. Immediate from the definition of $\bar{\mathcal{P}}_{\sim}$. ■

We now define a mapping i from $\mathcal{P}_{\sim}$ into $\bar{\mathcal{P}}_{\sim}$ by setting

$$i(\alpha) = \text{cl}(P \times Q), \tag{5.9}$$

where $\text{cl}(P) = \alpha$ and $\text{cl}(Q) = 0$, the identity in $\mathcal{P}_{\sim}$.

PROPOSITION 5.10. *The mapping $i: \mathcal{P}_{\sim} \rightarrow \bar{\mathcal{P}}_{\sim}$ is an embedding of semigroups.*

Proof. It is trivial to check that i is well defined, injective, and $i(\alpha + \beta) = i(\alpha) + i(\beta)$ for $\alpha, \beta \in \mathcal{P}_{\sim}$. ■

We will denote the class of $P \times Q$ in $\bar{\mathcal{P}}_{\sim}$ by $\text{cl}(P) - \text{cl}(Q)$. Now the mappings $\alpha \rightarrow X^{\alpha} \in C(\mathcal{P}_{\sim})$ and $\alpha - \gamma \rightarrow X^{\alpha - \gamma} \in C(\bar{\mathcal{P}}_{\sim})$ are one-to-one correspondences between $\mathcal{P}_{\sim}$, $\bar{\mathcal{P}}_{\sim}$ and bases of $C(\mathcal{P}_{\sim})$ and $C(\bar{\mathcal{P}}_{\sim})$, respectively. Thus the embedding $i: \mathcal{P}_{\sim} \rightarrow \bar{\mathcal{P}}_{\sim}$ lifts to a unique map $\iota: C(\mathcal{P}_{\sim}) \rightarrow C(\bar{\mathcal{P}}_{\sim})$, where

$$\iota(X^{\alpha}) = X^{i(\alpha)}. \tag{5.11}$$

PROPOSITION 5.12. *The mapping $\iota: C(\mathcal{P}_\sim) \rightarrow C(\bar{\mathcal{P}}_\sim)$ is an injective bialgebra map.*

Proof. Trivial.

We shall denote $C(\bar{\mathcal{P}}_\sim)$ by $\bar{C}(\mathcal{P}_\sim)$ and this Hopf algebra will be called the *reduced incidence Hopf algebra of $\mathcal{P} \bmod \sim$* .

Recall that the set of multiplicative functions $M(H)$ on a Hopf algebra H forms a group under convolution. We define a map $f \rightarrow \bar{f}$ from $M(\bar{C}(\mathcal{P}_\sim))$ to $M(C(\mathcal{P}_\sim))$ by

$$\bar{f} = f \circ \iota. \quad (5.13)$$

PROPOSITION 5.14. *The mapping $f \rightarrow \bar{f}$ defines an isomorphism from the group of multiplicative functions on $\bar{C}(\mathcal{P}_\sim)$ to the group of multiplicative functions on $C(\mathcal{P}_\sim)$ which have convolution inverses.*

Proof. Immediate from the fact that ι is a coalgebra map. ■

In the special case when we start with a free incidence bialgebra $B(\mathcal{P}_\sim)$, the Hopf algebra $\bar{B}(\mathcal{P}_\sim)$ is called the *free incidence Hopf algebra of $\mathcal{P}$ modulo $\sim$* , and is denoted $H(\mathcal{P}_\sim)$. We then have injections

$$C(\mathcal{P}_\sim) \xrightarrow{\iota_1} B(\mathcal{P}_\sim) \xrightarrow{\iota_2} H(\mathcal{P}_\sim)$$

and a mapping $\psi: M(H(\mathcal{P}_\sim)) \rightarrow C^*(\mathcal{P}_\sim)$ given by $\psi(f) = f \circ \iota_1 \circ \iota_2$ for $f \in M(H(\mathcal{P}_\sim))$.

PROPOSITION 5.15. *The map ψ is an isomorphism from the group of multiplicative functions on $H(\mathcal{P}_\sim)$ onto the group of invertible elements of $C^*(\mathcal{P}_\sim)$.*

Proof. Immediate from Propositions 3.7 and 5.14. ■

EXAMPLE 5.16 (The inverse of an ordinary power series.) Let $\mathcal{P}$ be the family of all finite linearly ordered sets, and let $\sim$ be the isomorphism relation. Then types correspond bijectively with non-negative integers by setting n equal to the class of an $n+1$ element linearly ordered set, for $n \geq 0$. The incidence coalgebra $C(\mathcal{P}_\sim)$ has basis X^n , $n \geq 0$. And if for $f \in C^*(\mathcal{P}_\sim)$ we define $f(t)$ by

$$f(t) = \sum_{n \geq 0} f(X^n) t^n$$

then the mapping $f \rightarrow f(t)$ gives an isomorphism between the dual algebra $C^*(\mathcal{P}_\sim)$ and the algebra $K(t)$ of ordinary formal power series under

multiplication. The free incidence bialgebra $B(\mathcal{P}_\sim)$ is isomorphic to the polynomial algebra $K[X_0, X_1, X_2, \dots]$, where $\Delta(X_n) = \sum_{i=0}^n X_i \otimes X_{n-i}$. The free incidence Hopf algebra $H(\mathcal{P}_\sim)$ is isomorphic to $K[X_0, X_1, X_2, \dots]$ localized at X_0 , with $\Delta(X_n) = \sum_{i=0}^n X_i \otimes X_{n-i}$ for $i \geq 0$ and $\Delta(1/X_0) = 1/X_0 \otimes 1/X_0$. It is easy to check (using expansion by cofactors) that the determinantal expression

$$S(X_n) = (-1) X_0^{-n-1} \begin{vmatrix} X_1 & X_2 & \cdots & X_n \\ X_0 & X_1 & & \vdots \\ 0 & X_0 & \ddots & \vdots \\ & 0 & \ddots & \vdots \\ & & 0 & X_0 & X_1 \end{vmatrix}$$

satisfies the recursive formula for the antipode given in Theorem 4.1. Hence if $f \in C^*(\mathcal{P}_\sim)$, $f(X^n) = a_n$ for $n \geq 0$, and $a_0 \neq 0$. Then the multiplicative function $\tilde{f}$ on $H(\mathcal{P}_\sim)$ corresponding to f by the map ψ in Proposition 5.15 is given by $\tilde{f}(X_n) = a_n$, $n \geq 0$ and $\tilde{f}(1/X_0) = 1/a_0$.

If $g(t) = \sum_{n \geq 0} b_n t^n$ is the multiplicative inverse of $f(t)$ we then have

$$b_n = \tilde{f} \circ S(X_n) = (-1)^{n-1} a_0^{-n-1} \begin{vmatrix} a_1 & a_2 & \cdots & a_n \\ a_0 & a_1 & & \vdots \\ & 0 & \ddots & \vdots \\ & & 0 & a_0 & a_1 \end{vmatrix},$$

which is the classical determinantal formula for $f^{-1}(t)$ (see [2, p. 157]).

VI. A CLOSED FORMULA FOR THE ANTIPODE

Let $\mathcal{P}_\sim$ be a family of types with a commutative semigroup structure, denoted by $+$, such that the reduced incidence coalgebra $C(\mathcal{P}_\sim)$ becomes a (commutative) Hopf algebra with the induced multiplication. Let P be a segment of type α in $\mathcal{P}_\sim$. If C is a chain $\hat{0} = \sigma_0 < \sigma_1 < \cdots < \sigma_n = \hat{1}$ in P then the *length* of C , $l(C)$, is defined to be n , the number of segments in the chain. Also we define types $\Omega(C)$ and $\Omega_0(C)$ in $\mathcal{P}_\sim$ as

$$\Omega(C) = \sum_{i=1}^n \text{cl}([\sigma_{i-1}, \sigma_i])$$

and

$$\Omega_0(C) = \sum_{i=0}^n \text{cl}([\sigma_i, \sigma_i]).$$

Using these notations we can now give an explicit formula for the antipode S of $C(\mathcal{P}_\sim)$.

Theorem 6.1. *The antipode S of $C(\mathcal{P}_\sim)$ is given by*

$$S(X^\alpha) = \sum_C (-1)^{l(C)} X^{\Omega(C) - \Omega_0(C)}, \quad (6.2)$$

where the sum ranges over all chains C from $\hat{0}$ to $\hat{1}$ in a segment of type α in $\mathcal{P}_\sim$.

Proof. If α is a one-point type then there is only one (trivial) chain C in a segment of type α , and we have $l(C) = 0$, $\Omega(C) = 0$, and $\Omega_0(C) = \alpha$. So (6.2) gives $S(X^\alpha) = X^{-\alpha}$ and thus $S * I(X^\alpha) = I * S(X^\alpha) = e$, the unit of $C(\mathcal{P}_\sim)$. If α is not a one-point type let $P \in \alpha$. Then

$$\begin{aligned} S * I(X^\alpha) &= \sum_{\sigma \in P} S(X^{\sigma^-}) X^{\sigma^+} \\ &= S(X^\alpha) X^{\hat{1}^+} + \sum_{\substack{\sigma \in P \\ \sigma < \hat{1}}} S(X^{\sigma^-}) X^{\sigma^+}. \end{aligned}$$

Using formula (6.2), we rewrite this sum as

$$\begin{aligned} &S(X^\alpha) X^{\hat{1}^+} + \sum_{\substack{\sigma \in P \\ \sigma < \hat{1}}} \sum_{C' \text{ chain in } [\hat{0}, \sigma]} (-1)^{l(C')} X^{\Omega(C') - \Omega_0(C') + \sigma^+} \\ &= S(X^\alpha) X^{\hat{1}^+} + \sum_{C' \text{ chain in } P} (-1)^{l(C')-1} X^{\Omega(C') - (\Omega_0(C') - \text{cl}([\hat{1}, \hat{1}]))} \\ &= S(X^\alpha) X^{\hat{1}^+} - \left(\sum_{C' \text{ chain in } P} (-1)^{l(C')} X^{\Omega(C') - \Omega_0(C')} \right) X^{\hat{1}^+}. \end{aligned}$$

Now if we apply formula (6.2) to evaluate $S(X^\alpha)$ in the above, we conclude that $S * I(X^\alpha) = 0$ when we use (6.2) to define S . Similarly we find that $I * S(X^\alpha) = 0$. Thus the theorem is proved. ■

EXAMPLE 6.3 (The inverse of a Dirichlet series). Let $\mathcal{P}$ consist of the single poset $\mathbb{Z}^+$ of positive integers ordered by divisibility. Set $[x, y] \sim [u, v]$ if $y/x = v/u$. Then the set of types $\mathcal{P}_\sim$ corresponds to the positive integers by setting $\text{cl}([x, y]) = y/x$. The free incidence bialgebra $B(\mathcal{P}_\sim)$ has a basis consisting of monomials in the variables $X_1, X_2, X_3, \dots$, and if f is a multiplicative function on $B(\mathcal{P}_\sim)$ we define a formal Dirichlet series $f(s)$ by

$$f(s) = \sum_{n \geq 1} \frac{f(X_n)}{n^s}.$$

The correspondence $f \rightarrow f(s)$ gives an isomorphism from the semigroup $M(B(\mathcal{P}_\sim))$ of multiplicative functions on $B(\mathcal{P}_\sim)$ under convolution onto the semigroup of formal Dirichlet series under Dirichlet convolution. Hence the group of multiplicative functions $M(H(\mathcal{P}_\sim))$ on the free incidence Hopf algebra $H(\mathcal{P}_\sim)$ is isomorphic to the group of Dirichlet series which have convolution inverses, i.e., $\{f(s) = \sum_{n \geq 1} (a_n/n^s) \mid a_1 \neq 0\}$. By Theorem 6.1, the antipode S of $H(\mathcal{P}_\sim)$ is given by

$$S(X_1) = 1/X_1$$

and

$$\begin{aligned} S(X_n) &= \sum_{C \text{ chain in } [1, n]} (-1)^{\ell(C)} X^{Q(C) - Q_0(C)} \\ &= \sum_{\substack{d_1 d_2 \cdots d_k = n \\ d_i \neq 1}} \frac{(-1)^k}{X_1^{k+1}} X_{d_1} X_{d_2} \cdots X_{d_k} \end{aligned}$$

for $n > 1$, where the sum is taken over all ordered factorizations $d_1 d_2 \cdots d_k = n$ of n with no $d_i = 1$. Now suppose $f(s) = \sum_{n \geq 1} (a_n/n^s)$, $a_1 \neq 0$, and $f^{-1}(s) = \sum_{n \geq 1} b_n/n^s$ is the inverse of $f(s)$ under Dirichlet convolution. Since inverses in $M(H(\mathcal{P}_\sim))$ are given by composing with the antipode S , we have immediately that

$$b_1 = 1/a_1$$

and

$$b_n = \sum_{\substack{d_1 d_2 \cdots d_k = n \\ d_i \neq 1}} (-1)^k a_{d_1} a_{d_2} \cdots a_{d_k} / a_1^{k+1}.$$

VII. THE FAÀ DI BRUNO HOPF ALGEBRA

Let $\mathcal{F}$ be the Faà di Bruno bialgebra, introduced in Example 1.11. Recall that $\mathcal{F}$ is the reduced incidence coalgebra $C(\mathcal{P}_\sim)$ of the family $\mathcal{P}$ of all lattices of partitions of finite sets modulo the relation $\sim$, given by $[\pi_1, \sigma_1] \sim [\pi_2, \sigma_2]$ if and only if $\text{cl}([\pi_1, \sigma_1]) = \text{cl}([\pi_2, \sigma_2])$, where $\text{cl}([\pi, \sigma]) = \alpha$ is the *type vector* $\alpha = (\alpha_1, \alpha_2, \dots)$ of the interval $[\pi, \sigma]$, in which α_i is equal to the number of blocks of σ which consist of exactly i blocks of π . As an algebra, $\mathcal{F}$ is isomorphic to the polynomial algebra $K[X_1, X_2, \dots]$, where the isomorphism is given by mapping X^α to the monomial $X_1^{\alpha_1} X_2^{\alpha_2} \cdots$.

The *Faà di Bruno Hopf algebra* $\mathcal{F}$ is the reduced incidence Hopf algebra

$\bar{C}(\mathcal{P}_\sim)$. It has the algebra structure of $K[X_1, X_2, \dots]$ localized at X_1 , and diagonalization and counit as in $\mathcal{F}$, with the addition of $\Delta(1/X_1) = 1/X_1 \otimes 1/X_1$ and $\varepsilon(1/X_1) = 1$.

Now let $M(\mathcal{F})$ denote the semigroup of multiplicative functions on $\mathcal{F}$. If $f \in M(\mathcal{F})$, define a formal power series $f(t)$ by

$$f(t) = \sum_{n \geq 1} f(X_n) t^n / n!. \quad (7.1)$$

It has been shown that the mapping $f \rightarrow f(t)$ is an anti-isomorphism from the semigroup $M(\mathcal{F})$ under convolution onto the semigroup of formal exponential power series with zero constant term under the operation of composition (see [4]).

Now suppose $f \in M(\bar{\mathcal{F}})$, the group of multiplicative functions on the Hopf algebra $\bar{\mathcal{F}}$, and $f(X_n) = a_n$ for $n \geq 1$. If we map $f \rightarrow f(t)$ by (7.1) then by Proposition 5.14, we have an anti-isomorphism from $M(\bar{\mathcal{F}})$ onto the group of invertible formal power series under composition. Hence if $g(t) = \sum_{n \geq 1} b_n t^n / n!$ is the compositional inverse of $f(t) = \sum_{n \geq 1} a_n t^n / n!$, then we have

$$b_n = f \circ S(X_n), \quad (7.2)$$

where S is the antipode of $\bar{\mathcal{F}}$.

One form of the classical Lagrange Inversion formula (see [2]) gives the coefficients of the inverse series g by

$$b_n = \sum_{k=1}^{n-1} (-1)^k (a_1)^{-n-k} B_{n+k-1}(0, a_2, a_3, \dots), \quad (7.3)$$

where the $B_{p,k}$ are the *exponential Bell polynomials* given by

$$B_{p,k}(X_1, X_2, \dots) = \sum_{\substack{\sigma \in \Pi_p \\ |\sigma| = k}} X^{\sigma^-} \quad \text{in } \bar{\mathcal{F}},$$

where the sum is over all partitions σ consisting of k blocks in the lattice Π_p of partitions of a p element set, and $X^{\sigma^-} = X_1^{\sigma_1} X_2^{\sigma_2} \cdots$ when $\text{cl}([\hat{0}, \sigma]) = (\sigma_1, \sigma_2, \dots)$.

Since f is multiplicative and $f(X_n) = a_n$, we can thus rewrite (7.2) as

$$b_n = f \left(\sum_{k=1}^{n-1} \sum_{\substack{\sigma \in \Pi_{n+k-1} \\ |\sigma| = k \\ \sigma_1 = 0}} \frac{(-1)^k}{X_1^{n+k}} X^{\sigma^-} \right) \quad (7.4)$$

(where the inner sum takes place over all k -block partitions σ in Π_{n+k-1} which have no blocks of size one). Thus we have proved the following:

THEOREM 7.5. *The antipode S of the Faà di Bruno Hopf algebra is given by*

$$S(X_n) = \sum_{k=1}^{n-1} \sum_{\substack{\sigma \in \Pi_{n+k-1} \\ |\sigma| = k \\ \sigma_1 = 0}} \frac{(-1)^k}{X_1^{n+k}} X^{\sigma^-}. \quad (7.6)$$

Proof. Compare Eqs. (7.2) and (7.4). ■

If we use the closed formula for the antipode, given by Theorem 6.1, to evaluate the left-hand side of (7.6) we then have the following combinatorial formulation of the Lagrange inversion formula (7.3):

COROLLARY 7.7. *In the Faà di Bruno Hopf algebra $\mathcal{F}$,*

$$\sum_{C \text{ chain in } \Pi_n} (-1)^{l(C)} X^{\Omega(C) - \Omega_0(C)} = \sum_{k=1}^{n-1} \sum_{\substack{\sigma \in \Pi_{n+k-1} \\ |\sigma| = k \\ \sigma_1 = 0}} \frac{(-1)^k}{X_1^{n+k}} X^{\sigma^-}, \quad (7.8)$$

where the sum on the left is taken over all chains C from $\hat{0}$ to $\hat{1}$ in the partition lattice Π_n .

Many cancellations occur in the sum on the left side of (7.8). This is because there are many pairs of chains C and C' in Π_n such that $\Omega(C) - \Omega_0(C) = \Omega(C') - \Omega_0(C')$, while $l(C)$ and $l(C')$ have different parity. On the other hand, the sum on the right is minimal in the sense that no cancellation of terms takes place. Thus the Lagrange inversion formula (7.8) can be viewed as a description of what remains of the alternating sum

$$\sum_{C \text{ chain in } \Pi_n} (-1)^{l(C)} X^{\Omega(C) - \Omega_0(C)}$$

after all cancellations are taken into account.

We believe that an understanding of exactly how these cancellations take place will not only provide a direct combinatorial proof of the Lagrange inversion formula, but may well yield analogous formulas for the antipodes of Hopf algebras arising from classes of geometric lattices other than partition lattices.

REFERENCES

1. G. BIRKHOFF, "Lattice Theory," 3rd ed., A.M.S. Colloquium Publications, Vol. XXV, Amer. Math. Soc., Providence, RI, 1967.
2. L. COMTET, "Advanced Combinatorics," Reidel, Dordrecht, 1974.
3. P. DOUBILET, "Studies in partitions and permutations," Ph.D. thesis, M.I.T., Cambridge, MA, 1973.

4. P. DOUBILET, G.-C. ROTA, AND R. STANLEY, On the foundations of combinatorial theory. VI. The idea of generating function, in "Sixth Berkeley Symposium on Mathematical Statistics and Probability," Vol. 2, pp. 267–318, Univ. of California Press, 1972.
5. ARNE DÜR, "Mobius Functions, Incidence Algebras and Power Series Representations," Lecture Notes in Mathematics, No. 1202, Springer-Verlag, Berlin/Heidelberg, 1986.
6. J. GOLDMAN AND G.-C. ROTA, On the foundations of combinatorial theory, IV. Finite vector spaces and Eulerian generating functions," *Studies in Appl. Math.*, Vol. 49, pp. 239–258, New York, 1970.
7. S. A. JONI AND G.-C. ROTA, Coalgebras and bialgebras in combinatorics, "Studies in Appl. Math.," Vol. 61, pp. 93–139, New York, 1979.
8. G.-C. ROTA, On the foundations of combinatorial theory. I. Theory of Möbius functions, *Z. Wahrsch. Verw. Gebiete* **2** (1964). 340–368.
9. D. A. SMITH, Incidence functions as generalized arithmetic functions I. *Duke Math. J.* **34** (1967), 617–634.
10. D. A. SMITH, Incidence functions as generalized arithmetic functions II, *Duke Math. J.* **36** (1969), 15–30.
11. D. A. SMITH, Incidence functions as generalized arithmetic functions III, *Duke Math. J.* **36** (1969), 343–368.
12. M. SWEEDLER, "Hopf Algebras," Benjamin, New York, 1969.